Group Project Briefing (G7)

Things to Note

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Synopsis

- To build a dynamic and interactive web application that utilizes the technologies you've learned this term including a back-end implementation for data storage and user authentication where necessary.

- The key learning outcome will be demonstrating your ability to integrate the various technologies you've studied and apply them effectively in your final solution.

Deliverables

- Progress Pitching (Week 9)
  - 5 minutes
- Web application (Week 12, Friday 9am)
  - Submission on eLearn
- Presentation (Week 13)
  - Live - 12 mins + 3 mins (Q&A)
  - Best practice video

Submission

- Deadline: Week 12 (Friday) 9am
- Late submission policy.

Submit within 1 hour of the specified deadline	10% penalty (of your score for the entire project)
Submit beyond 1 hour of the specified deadline	0 Mark

Submission

- A zip file consists of the following 3 items and submit via eLearn (Assignments --> Final Project Deliverables (GX)).

Web Application

- Your application must be located in the server's root folder, with 'index.html' serving as the entry page.

A README.txt file

- Include clear instructions on how to set up and run your application based on the submitted files.
- Include the landing page URL for your deployed and live application.
- Provide username/password details of any dummy account if it is needed to access your application.

Presentation Slides

- Your presentation slides (.pptx or .pdf).
- Include a link to your presentation video (best practice) on the title slide.

Minimum Requirements

- Web Application (General)
  - You must identify and solve a problem. This can be either an unsolved problem or an existing problem.
  - You must use HTML, CSS and JavaScript in your implementation. Optionally, you are encouraged to use Bootstrap (or similar CSS framework), JSON, and Vue (or a similar JavaScript framework).
  - Your solution must include a responsive design.
  - Your solution must be a dynamic site that interact with a backend data store. This could involve applying knowledge from IS113 (e.g. Express.js + MongoDB), but you are free to choose another data storage solution if you prefer (e.g. files, other types of databases, cloud storage).
  - Your solution must make meaningful use of at least one external publicly available API via asynchronous HTTP requests.

Minimum Requirements

- Web Application (General)
  - Your solution must include end-to-end (E2E) testing for all core features.
  - You must utilize GitHub collaboration tools available for your project to effectively manage version control, track changes, and facilitate communication among team members.
  - You must deploy your final solution to a remote server (e.g. AWS, Google Cloud, Azure, Firebase, Vercel and etc.) that is accessible online.

Minimum Requirements

- Presentation
  - Focus on the following:
    - On the solution:
      - What problem is your group trying to solve (supported by facts), and how is solving this problem useful?
      - Briefly describe how you solved the problem and met the project requirements.
        - Discuss the design of your solution.
        - Explain how you used JavaScript or similar frameworks to make your app dynamic and interactive.
        - Describe how you used CSS, Bootstrap, or similar CSS frameworks to style your app, ensuring that the web pages are responsive, clean (no irrelevant or extraneous information), and have clear, logical flows (intuitive to use).
      - Clearly state the type(s) of data store(s) you are using.
      - Identify the external APIs you have used and provide brief integration details.
      - Mention any technologies used beyond the minimum requirements (if applicable).

Minimum Requirements

- Presentation
  - Focus on the following:
    - On the presentation:
      - Ensure good flow and transitions.
      - Aim for ease of understanding.
      - Use concise and meaningful presentation materials (slides, videos, etc.).
      - Manage your time well:
        - Having a lot of extra time left with a presentation still unclear is not ideal.
  - Record your presentation (maximum 12 minutes, at least 720p - ideally 1080p) and upload it to YouTube. Please ensure that the video is either unlisted or public.

Assessment Components - Progress Pitching** (5%)

Component	Description	%
Problem Definition	Clearly articulates the problem being addressed, including goals and constraints.	25%
Concept Design & Technology Stack	Outlines the proposed solution, including design approach and chosen web technologies/frameworks.	25%
Task Delegation	Demonstrates effective team collaboration through clear distribution of roles and responsibilities.	25%
Feature Demonstration	Live demo of at least one working feature showcasing functionality and progress.	25%
		100%

***A detailed progress pitch deck will be released in Week 7.*

Assessment Components – Presentation and Web Apps (20%)

Component		%
(1) Presentation		25%
	The presentation	20%
	Q & A	5%
(2) Working web application		75%
	Problem(s) solving and solution appropriateness	20%
	Working app, correctly implemented and usable solution	27%
	Styling (including responsive design)	18%
	Testing	10%
		100%

Assessment Components – Peer Evaluation (5%)

- Peer Evaluation Submission:
  - Every team member must **submit 2 times of peer evaluation form (week 7 and week 11)**.
- Internal Conflict Resolution:
  - **Teams must actively manage and resolve conflicts internally** instead of waiting to report them on the peer evaluation form.
- Early Intervention:
  - **Report unmanageable team members to the teaching team by week 7**.
- Shared Responsibility:
  - If a team fails to attempt conflict resolution, all members may receive the same grade regardless of peer scores.
- Grading Criteria:
  - Teamwork and project management are implicit components of the final project grade.
- Severe Penalties:
  - **Members who fail to contribute significantly or miss most meetings face strict penalties**, up to a zero mark for the entire project.

Marking Rubrics (Progress Pitch)

Marks	Details
C+ & below	Problems / features / job scopes / web design / tech stacks are vague. There is no functioning task.
B- to B+	Problems / features / job scopes / web design / tech stacks are not very clear but somewhat defined. A task is somewhat functioning.
A- to A	Problems / features / job scopes / web design / tech stacks are clearly defined. A task is functioning well.
A+	Challenging Problems / features / job scopes / web design / tech stacks are clearly defined. A challenging task is functioning well.

Marking Rubrics (Presentation)

Marks	Details
C+ & below	**Presentation with Demo is sloppy.** • Difficult to understand how the solution addresses the problem, or the clarity of the technologies used.
B- to B+	**Presentation with Demo is of reasonable standard.** • Relatively clear presentation and demo that shows how technologies are used to address the problem with some minor hiccups.
A- to A	**Presentation with Demo is good.** • Clear presentation and demo that are quite seamless. Easy to understand, with a clear explanation of what was done and how it was accomplished.
A+	**Presentation with Demo is exceptional.** • Crystal-clear presentation and demo that are seamless. No clarification questions are needed to understand what was done. Additionally, at least one bonus point must be earned to qualify.

Marking Rubrics (Web Application)

Problem(s) solving and solution appropriateness

Marks	Details
C+ & below	**Problem is not solved and/or not a useful app.**
B- to B+	**Problem is somewhat solved and is relatively useful.** • The problem might not be completely solved but is mostly addressed.
A- to A	**Problem solved is very useful.** • A clearly defined problem that is addressed with a satisfactory solution.
A+	**Challenging problem is solved and very useful.** • It is a challenging problem to address, and the solution is highly appropriate.

Marking Rubrics (Web Application)

Working app, correctly implemented, usable solution[1]

Marks	Details
C+ & below	**App is somewhat working to not working.** • Intermittently working, with some parts functioning or completely non-functional.
B- to B+	**App is buggy with some working parts.** • Contains bugs, but none are critical or show-stopping. The app is roughly usable without requiring excessive effort to understand how to use it.
A- to A	**App is working well, and usability is good.** • Some minor bugs, but the app is relatively intuitive to use.
A+	**App is working perfectly with great usability.** • No bugs at all; the app is completely efficient and intuitive, requiring no effort to figure out how to use it.

[1] In the sense of the following guidelines: https://www.interaction-design.org/literature/article/user-interface-design-guidelines-10-rules-of-thumb

Marking Rubrics (Web Application)

Styling (including responsiveness)

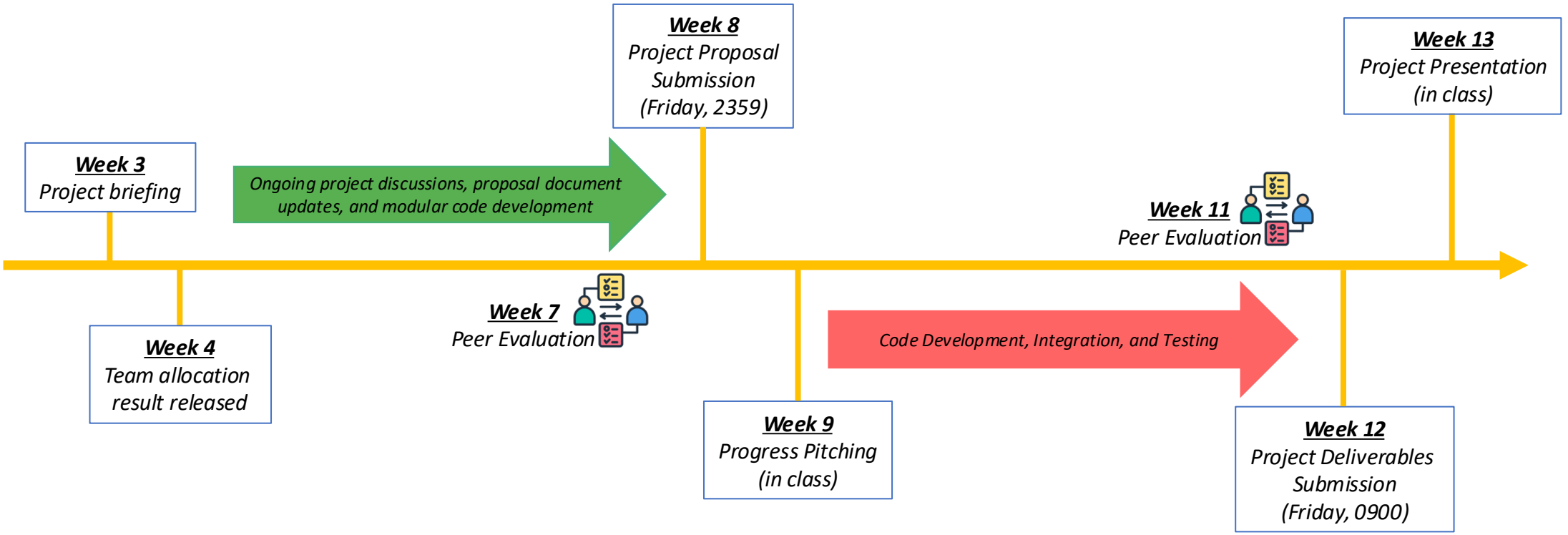
Marks	Details
C+ & below	**Limited styling implemented; Minimal responsiveness implemented.**
B- to B+	**Relatively decent styling and somewhat responsive site** • Somewhat clean and appealing, and the site is somewhat responsive, with minor issues but nothing critical.
A- to A	**Good style and somewhat responsive site** • Attractive and well-executed, and the site is fully responsive.
A+	**Excellent style and fully responsive site** • A beautiful site that makes effective use of style and is fully responsive, scaling appropriately for different screen sizes and orientations.

Marking Rubrics (Web Application)

Testing

Marks	Details
C+ & below	**Limited coverage of core features.** • Tests are limited, generally non-repeatable and not stable. Provide very limited instructions in README on how to run tests.
B- to B+	**Decent coverage of core features** • Tests cover some of the app's main user journeys. Some of the tests are repeatable and stable. Provide some brief instructions in README on how to run tests. Tests demonstrate somewhat understanding of testing concepts.
A- to A	**Good coverage of core features with some quality test cases** • Tests cover most of the app's main user journeys. Tests are mostly repeatable, generally stable and have clear structure. Provide somewhat clear instructions in README on how to run tests. Tests demonstrate a good understanding of testing concepts.
A+	**Excellent coverage of core features with quality test cases** • Tests cover the app's main user journeys. Tests are repeatable and use stable selectors and have clear test structure. Provide clear instructions in README on how to run tests. Tests demonstrate excellent understanding of testing concepts.

Timeline


Week 3
Project briefing
Week 4
Team allocation result released
Ongoing project discussions, proposal document updates, and modular code development
Week 7
Peer Evaluation
Week 8
Project Proposal Submission (Friday, 2359)
Week 9
Progress Pitching (in class)
Code Development, Integration, and Testing
Week 11
Peer Evaluation
Week 12
Project Deliverables Submission (Friday, 0900)
Week 13
Project Presentation (in class)

Action Items

- Fill up the form https://forms.gle/Cp7bsp4cyWTNKSL7A to facilitate the group allocation process. Each group will be allocated 5-6 members.
- Appoint a leader for your team to guide and manage the project effectively.
- Leaders to download the Project Proposal Template from eLearn and upload it to your Google Drive. Then, share the document with the rest of the member and ccwong@smu.edu.sg
- The team should begin meeting, brainstorming, and discussing the project. Document all your discussions in the project proposal.
- The team should finalize the project idea and conduct a final review of the project proposal document before the submission deadline on Sunday of Week 8 at 23:59 via merged eLearn (Assignments --> Project Proposal (GX)).

Common Project Ideas

- Event Management Portal
- Donation Drive Portal
- Recipe/Meals Finder Portal
- Job Finder Portal
- Matchmaking Portal
- Pet Adoption Portal
- Volunteer Recruitment Portal

Don't start with
"What portal can we build?"

Start with
"What interesting problem can we solve?"

You may still propose these topics, but ***we strongly encourage you to explore a less common problem domain or bring a genuinely new perspective to an existing one***.

Frequently Asked Questions (FAQ)

- Are we required to use Bootstrap or Vue.js?
  - No, you are not required to use Bootstrap or Vue.js specifically. However, you are encouraged to use frameworks to simplify, standardize, and speed up the development process of your project.
- I want to use React. Is this allowed?
  - Yes, using React or a similar framework is allowed. However, we strongly encourage you to use Vue.js for the following reasons:
    - Technical Support: Vue.js is covered in the syllabus, so we can provide more effective support. React is not officially included in the course materials.
    - Exam Preparation: The final exam will include questions on JavaScript and Vue.js, but not on React. Using Vue.js will better prepare you for the assessment.
  - Also, consider your teammates' familiarity with React. If everyone is comfortable with it, using React is fine. However, if some team members are not, they would need to learn two different JavaScript frameworks during the course. Make sure the whole team is comfortable with this decision.

Frequently Asked Questions (FAQ)

- Can my instructional team provide technical support on on the technology we are using, e.g., Docker, AWS, etc.?
  - In general, no. There is a plethora of technologies out there. Given the variability in our expertise with different technologies, it would be unfair for us to assist one team with technology we are familiar with while being unable to support another team with technology, we are not knowledgeable about. However, we are happy to engage in high-level discussions regarding the appropriateness of different technologies for your solution.

- Which browser should we use?
  - You can use any browser you prefer. However, for grading purposes, please ensure that your website displays properly in Google Chrome. We will also use Chrome's developer tools to test your site's responsive design.

Frequently Asked Questions (FAQ)

- Are we allowed to use publicly available code sources (e.g. from GitHub projects)?
  - You may use publicly and freely available libraries (i.e., not paid). However, such usage must be clearly identified in your presentation and within the comments of your code. You must NOT present this code as your own work, as that would constitute plagiarism. If you are unsure, please contact your instructional team to discuss your plans.
- Are all members of a group receiving the same group project grade?
  - Yes, a single grade is awarded for the project. However, if a group member is not participating, please inform your instructors EARLY, and individual adjustments will be considered on a case-by-case basis. However, if you notify your instructors too close to the final deliverable deadline (e.g., one week before), it may be considered too late to take any meaningful action, and your instructors may not accept your case.

Frequently Asked Questions (FAQ)

- Are all members of a group required to code?
  - Yes, all members should contribute to the coding of the project. It is advisable to assign specific tasks to each member (e.g., Bob handles styling, and Olivia manages scripting) to ensure effective collaboration.
- Are all members of a group required to present?
  - Yes, all members should take this opportunity to practice their presentation skills.
- Are we allowed to use back-end code from IS113 and build on it?
  - Yes, you are allowed to build upon example back-end code from IS113. In IS216, we will provide an example of how to apply styling and scripting to some back-end code, and you are welcome to build on that as well if you wish.

Frequently Asked Questions (FAQ)

- Will you be grading the backend code?
  - While there is no explicit grading component specifically for the back-end code, the overall functionality of your application is evaluated. Therefore, the quality of your back-end implementation will be assessed under the grading criterion for a “working app, correctly implemented, and usable solution.”
- If other groups are tackling the same problem, would we be penalized?
  - No. There are often many ways to solve the same problem, and multiple groups working on similar issues is acceptable.

Frequently Asked Questions (FAQ)

- Can I use dummy data instead of real-data?
    - If the type of data your application requires is publicly available, we encourage you to use real data to make your prototype more realistic. However, if you anticipate difficulties in obtaining real data, please consult with your instructional team for guidance.
- Under “problem solving and solution appropriateness”, are you referring to coding complexity, or more to the business side?
    - We are referring to the business side, meaning you should clearly articulate the problem your website addresses and demonstrate that the implementation effectively solves this problem. The problem does not need to be a critical societal issue. For example, a virtual escape room may not address a major societal problem but still solves a relevant issue, such as enabling remote connection with friends and family, especially during a pandemic.

Frequently Asked Questions (FAQ)

- Are we allowed to scrape data from a website?
  - If you choose to scrape data from a website, you must ensure that you comply with any restrictions outlined in the site's robots.txt file and consider any legal implications. Additionally, you must clearly identify and properly credit the sources of your data. Note that using scraped data does not substitute the requirement to utilize at least one publicly available API.
- How will the responsive design of our web application be tested?
  - We will test your website using Chrome's developer tools across various viewports, starting from the iPhone 6 to the XL breakpoint of Bootstrap. Ensure that your layout looks good and adjusts properly across all viewports within this range.

Frequently Asked Questions (FAQ)

- The API I'm using is free, but only up to certain amounts of usage. How much testing will you be doing after submission?
    - As a rough estimate, each functionality will be tested up to 5-6 times per day. You will need to assess whether this falls within the free usage limits of your API.

Frequently Asked Questions (FAQ)

- Are we allowed to use AI/LLM in the project?
  - Generally, we allow the use of AI/LLM for lower order thinking tasks such as:
    - information search
    - Generating website concepts, layouts, or themes
    - Exploring different UI/UX design inspirations
    - Explaining coding errors and debugging hints
    - Boilerplate code generation – starter code and code snippets
    - Generating unit test cases, sample inputs, or mock data
  - You are not allowed to use AI/LLM for higher order thinking tasks such as:
    - Core implementation tasks (to implement main business logic, backend endpoints, or critical frontend interactivity)
    - Problem solving – students must demonstrate their own capability for solving problems and addressing major implementation issues, not just rely on automated AI fixes.
  - In the README.txt file, state clearly the parts you solely rely on AI/LLM.

Exemplar Project (FLIP) Demo

https://is216-flip.vercel.app

https://github.com/cheinwongsmu/is216-flip

End of Briefing

Questions?